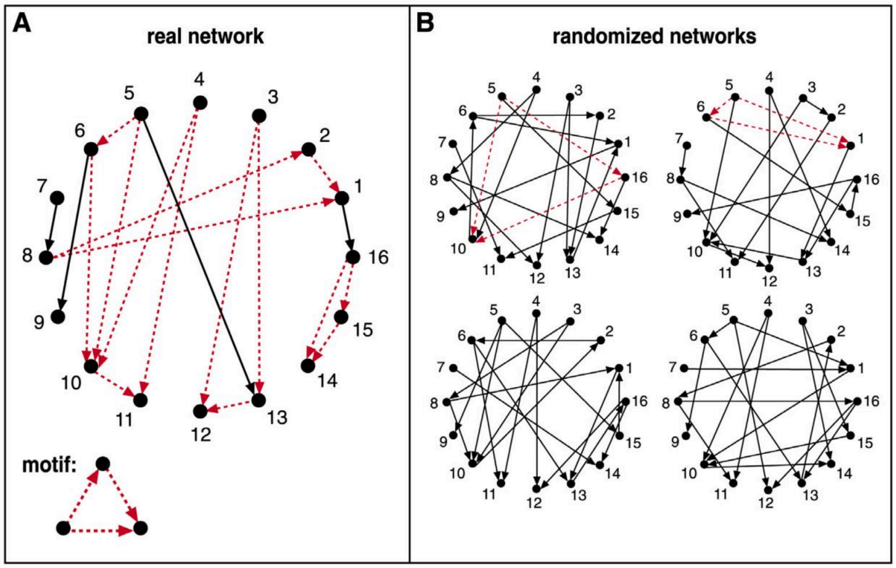
L5: Statistical Test For The Frequency of a Network Motif



Suppose we are given the 16-node network *G* at the left, and we want to examine if the FFL network motif *(shown at the bottom left)* appears ***too frequently***. How would we answer this question?

First, we need to count how many distinct instances of the FFL motif appear in *G*. One way to do so is to go through each node *u* that has at least two outgoing edges. For all distinct pairs of nodes *v* and *w* that *u* connects to uni-directionally, we then check whether *v* and *w* are also connected with a uni-directional edge. If that is the case *(u,v,w)* is an FFL instance. Suppose that the count of all FFL instances in the network *G* is *m(G)*.

We then ask: how many times would the FFL motif take place in a randomly generated network G_r that **a)** has the same number of nodes with *G*, and **b)** each node *u* in G_r has the same in-degree and out-degree with the corresponding node *u* in *G*? One way to create G_r is to start from *G* and then ***randomly rewire*** it as follows:

- Pick a random pair of edges of *(u,v)* and *(w,z)*
- Rewire them to form two new edges *(u,z)* and *(w,v)*
- Repeat the previous two steps a large number of times relative to the number of edges in *G*.

Note that the previous rewiring process generates a network G_r that preserves the in-degree and out-degree of every node in *G*. We can now count the number of FFL instances in G_r – let us call this count $m(G_r)$.

The previous process can be repeated for many randomly rewired networks G_r *(say 1000 of them).* This will give us an ensemble of networks G_r. We can use the counts $m(G_r)$ to form an empirical distribution of the number of FFL instances that would be expected by chance in networks that have the same number of nodes and the same in-degree and out-degree sequences as *G*.

We can then compare the count $m(G)$ of the given network with the previous empirical distribution to estimate the probability with which the random variable $m(G_r)$ is larger than $m(G)$ in the ensemble of randomized networks. If that probability is very small (say less than 1%) we can have a 99% statistical confidence that the FFL motif is much more common in *G* than expected by chance.

Similarly, if the probability with which the random variable $m(G_r)$ is smaller than $m(G)$ is less than 1%, we can have 99% confidence that the FFL motif is much less common in *G* than expected by chance. The magnitude of $m(G)$ relative to the average plus *(or minus)* a standard deviation of the distribution $m(G_r)$ is also useful in quantifying how common a network motif is.

The method we described here is only a special case of the general ***bootstrapping*** method in statistics.